

```
$ node breakfast.js --mode=async
```

JavaScript □□□

```
// 00000000 Promise 0 async/await
```

sync

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promise

□□□□□

await

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event loop

□□□□

\$ cat menu.md



01 /



vs □□□□□

02 / Promise

□□□□□then/catch□□□□□

03 / async & await



vs □□

04 / Event Loop

VIP □□□□□□□□□

🚧🚧🚧🚧🚧🚧 async □□□□□□□□□□□□

PART 01



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```
1  function makeCoffee() {
2      console.log("☕☕☕☕...");
3      const start = Date.now();
4      while (Date.now() - start < 2000) {} // ☐☐☐
5      console.log("☕☕☕☕");
6  }
7
8  function prepareBreakfastSync() {
9      makeCoffee(); // 2s
10     makeToast(); // 1.5s
11     fryEgg();     // 1s
12 }
```

CONSOLE — 4.5s

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PART 02

Promise

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pending

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fulfilled

□□□□resolve("🍌🍌")

rejected

□□□□reject("🍌🍌")

```
1 let myPromise = new Promise((resolve, reject) => {
2   let success = true;
3   setTimeout(() => {
4     if (success) resolve("🍌🍌");
5     else reject("🍌🍌");
6   }, 2000);
7 });
8
9 myPromise.then(result => console.log(result))
10 .catch(error => console.log(error));
```

□□□□□Promise.race + AbortController

```
1  const delay = (ms) => new Promise((resolve) => setTimeout(resolve, ms));
2
3  const fetchWithTimeout = (url, { timeout = 3000 } = {}) => {
4    const controller = new AbortController();
5    const timer = delay(timeout).then(() => controller.abort());
6
7    const request = fetch(url, { signal: controller.signal });
8
9    return Promise.race([request, timer]).catch((error) => {
10      if (error.name === "AbortError") return Promise.reject("□□□□");
11      return Promise.reject(error.message || "□□□□");
12    });
13  };
```

Promise.race

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AbortController

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PART 03

async / await

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then → await

```
1 async function loadSpells() {
2   try {
3     const res = await fetch("https://hp-api.onrender.com/api/spells");
4     const data = await res.json();
5     console.log("spells", data.length);
6   } catch (err) {
7     console.error(err.message);
8   }
9 }
```

👉👉👉👉👉 await 👉 then 👉👉👉👉👉👉👉👉👉👉👉 async/await👉👉👉👉 then👉



```
1  async function serial() {  
2    const coffee = await makeCoffee();  
3    const toast = await makeToast();  
4    return [coffee, toast];  
5  }
```

TOTAL TIME

$\approx 2 + 1.5 = 3.5s$

□□

token API — □□□□□



```
1  async function parallel() {  
2    const [coffee, toast] = await Promise.all([  
3      makeCoffee(),  
4      makeToast(),  
5    ]);  
6    return [coffee, toast];  
7  }
```

TOTAL TIME

$\approx \max(2, 1.5) = 2s$

□□

token API

Promise.all vs allSettled

	Promise.all	Promise.allSettled
□□□□□		□□□□□□□
□□	□□□□	{status, value/reason} □□
□□	□□□□□□□	□□□□□□□□□□

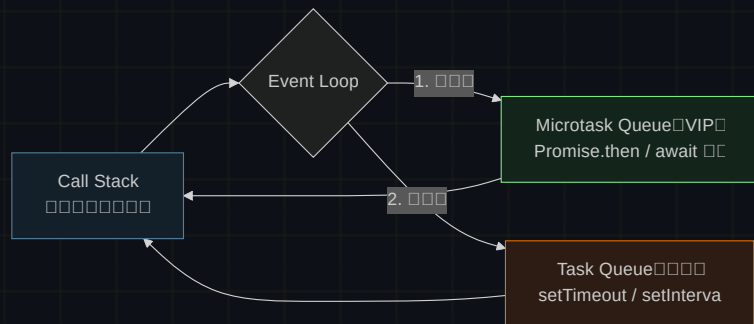
\$ npm install all allSettled

PART 04

Event Loop

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```
1 console.log("1");
2 setTimeout(() => console.log("2"), 0);
3 Promise.resolve().then(() => console.log("3"));
4 console.log("4");
```

ANSWER

1 → 4 → Call Stack

→ 3 → VIP

→ 2 → 0

1 → 4 → 3 → 2

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□□□□□ pending → fulfilled / rejected

await Promise.all

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Breakfast served.

fetch AbortController

```
$ node breakfast.js --mode=async
```

promise → await → race → event loop